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DETAIL SPECIFICATION

VALVE ACTUATOR, DIRECT COUPLED, GEAR-DRIVEN ELECTRO-MECHANICAL FOR NAVAL SHIPBOARD PROPULSION AND AUXILIARY SYSTEMS

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers electro-mechanical, gear-driven, direct coupled valve actuators (see 6.3.1) for fluid systems located throughout naval surface ships in shipboard propulsion, machinery, and auxiliary systems. An actuator's function is to provide enough torque to operate an attached valve to a required position.

1.2 Classification. Actuators are of the following types, classes, services, and travel, as specified (see 6.2).

1.2.1 Types.

a. Type 1 – Linear motion valves (gate, globe)

b. Type 2 – Rotational motion valves (ball, butterfly, plug)

1.2.2 Classes.

a. Class 1 – General application

b. Class 2 – Corrosive environment

c. Class 3 – Light-weight application

1.2.3 Services.

a. Service 1 – Submersible

b. Service 2 – Explosion proof

c. Service 3 – Submersible and explosion proof

1.2.4 Travel.

a. Travel 1 – Frequent full open and full shut duty cycle (see 6.3.4) operations

b. Travel 2 – Infrequent full open and full shut duty cycle operations

c. Travel 3 – Modulating operations - Meant for control valve applications where the valve may have to be placed in a certain position or percentage to aid in an operation that does not require a full open or full shut valve

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AMSC N/A FSC 4810

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d. Travel 4 – Frequent modulating operations - Meant for control valve applications where the valve is constantly adjusting the valve stem accordingly (i.e., Proportional-Integral-Derivative controls). These adjustments are not necessarily full open and full shut travels. They can be small increments or large valve movements.

1.3 Part or identifying number (PIN). PINs to be used for actuators acquired to this specification are created as follows:

M	**24657**	**-**	**X**	**X**	**X**	**X**
Prefix for Military Specification	Specification Number	-	Type	Class	Service	Travel
Example: M24657-1131 is valve actuator, linear motion, general application, submersible and explosion proof enclosures, and frequent OPEN/SHUT applications.						

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

QQ-N-286 - Nickel-Copper-Aluminum Alloy, Wrought (UNS N05500)

TT-P-1757 - Primer Coating, Alkyd Base, One Component

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-901 - Shock Tests, H.I. (High-Impact) Shipboard Machinery, Equipment, and Systems, Requirements for

MIL-DTL-917 - Electric Power Equipment, Basic Requirements for

MIL-DTL-2036 - Enclosures for Electric and Electronic Equipment, Naval Shipboard

MIL-DTL-2212 - Contactors and Controllers, Electric Motor AC or DC, and Associated Switching Devices

MIL-DTL-3661 - Lampholders, Indicator Lights, Indicator Light Housings, and Indicator Light Lenses, General Specification for

MIL-DTL-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5

MIL-PRF-8625 - Anodic Coatings for Aluminum and Aluminum Alloys

MIL-DTL-15090 - Enamel, Equipment, Light Gray (Navy Formula No. 111)

MIL-DTL-16884 - Fuel, Naval Distillate

MIL-M-17059 - Motors, 60-Cycle, Alternating-Current, Fractional H.P. (Shipboard Use)

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MIL-DTL-17060	-	Motors, Alternating Current, Integral-Horsepower, Shipboard Use
MIL-C-24368	-	Connector Assemblies; Plugs and Receptacles, Electric Power Transfer, Shore to Ship and Ship to Ship, General Specification for
MIL-DTL-24441	-	Paint, Epoxy-Polyamide, General Specification for
MIL-B-24480	-	Bronze, Nickel-Aluminum (UNS No. C95800) Castings for Seawater Service
MIL-PRF-24712	-	Coatings, Powder, Thermosetting
MIL-DTL-26482	-	Connectors, Electrical, Circular, Miniature, Quick Disconnect, Environment Resisting, Receptacles and Plugs, General Specification for
MIL-DTL-32613	-	Controller, Auxiliary-System, Naval Shipboard Use

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-167-1	-	Mechanical Vibrations of Shipboard Equipment (Type I – Environmental and Type II – Internally Excited)
MIL-STD-461	-	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
MIL-STD-810	-	Environmental Engineering Considerations and Laboratory Tests
MIL-STD-1310	-	Shipboard Bonding, Grounding, and Other Techniques for Electromagnetic Compatibility, Electromagnetic Pulse (EMP) Mitigation, and Safety
DOD-STD-1399-070-1	-	D.C. Magnetic Field Environment (Metric)
MIL-STD-1399-300-1	-	Low Voltage Electric Power, Alternating Current
MIL-STD-1399-502	-	Electronic Systems Parameters
MIL-STD-1472	-	Human Engineering
MIL-STD-1474	-	Noise Limits

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

NAVAL SEA SYSTEMS COMMAND (NAVSEA) PUBLICATIONS

T9070-AL-DPC-020/077-2	-	NAVSEA Hazardous Material Avoidance Process

(Copies of this document are available online via Technical Data Management Information System (TDMIS) at https://mercury.tdmis.navy.mil by searching for the document number without the suffix. Refer questions, inquiries, or problems to: DSN 296-0669, Commercial (805) 228-0669. This document is available for ordering (hard copy) via the Naval Logistics Library (NLL) at https://nll.navsup.navy.mil. For questions regarding the NLL, contact the NLL Customer Service at nllhelpdesk@navy.mil, (866) 817-3130, or (215) 697-2626/DSN 442-2626.)

U.S. GENERAL SERVICES ADMINISTRATION

GSA Global Supply Catalog

(Copies of this document are available online at www.gsaglobalsupply.gsa.gov.)

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AEROSPACE INDUSTRIES ASSOCIATION (AIA)

NAS411-1 - Hazardous Materials Target List

(Copies of this document are available online at www.aia-aerospace.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B1.1 - Unified Inch Screw Threads (UN, UNR, and UNJ Thread Forms)

(Copies of this document are available online at www.asme.org.)

ASTM INTERNATIONAL

ASTM A108	-	Standard Specification for Steel Bar, Carbon and Alloy, Cold-Finished
ASTM A269/A269M	-	Standard Specification for Seamless and Welded Austenitic Stainless Steel Tubing for General Service
ASTM A536	-	Standard Specification for Ductile Iron Castings
ASTM B148	-	Standard Specification for Aluminum-Bronze Sand Castings
ASTM B209/B209M	-	Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B283/B283M	-	Standard Specification for Copper and Copper-Alloy Die Forgings (Hot-Pressed)
ASTM B584	-	Standard Specification for Copper Alloy Sand Castings for General Applications
ASTM D395	-	Standard Test Methods for Rubber Property – Compression Set
ASTM D2000	-	Standard Classification System for Rubber Products in Automotive Applications
ASTM D4894	-	Standard Specification for Polytetrafluoroethylene (PTFE) Granular Molding and Ram Extrusion Materials

(Copies of these documents are available online at www.astm.org.)

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION (ISO)

ISO/IEC 17025 - General Requirements for the Competence of Testing and Calibration Laboratories

(Copies of this document are available online at www.iso.org.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

ANSI/NEMA 250 - Enclosures for Electrical Equipment (1,000 Volts Maximum)

(Copies of this document are available online at www.nema.org.)

SAE INTERNATIONAL

SAE ARP1481 - Corrosion Control and Electrical Conductivity in Enclosure Design

SAE AS50151 - Connectors, Electrical, Circular Threaded, AN Type, General Specification for

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SAE AMS7259 - Rubber: Fluorocarbon (FKM) High Temperature/Fluid Resistant Low Compression Set/85 to 95 Hardness for Seals in Fuel Systems and Specific Engine Oil Systems

SAE AMS7276 - Rubber: Fluorocarbon (FKM) High Temperature – Fluid Resistant Low Compression Set/70 to 80 Type A Hardness for Seals in Fuel Systems and Specific Engine Oil Systems

(Copies of these documents are available online at www.sae.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.2 Materials. Materials used for electro-mechanical valve actuators shall meet the requirements specified in table I. Materials that are susceptible to corrosion, rust, oxidation, or any other type of performance or material degradation shall be coated. Material for components not specifically detailed in table I shall be selected to prevent galling, seizing, and excessive wear of operating parts and galvanic compatibility with mating parts (see 4.5.1).

TABLE I. Electro-mechanical valve actuator material requirements.

Component	Class 1 and 3	Class 2
Enclosure	304 corrosion resistant steel (CRES) (ASTM A269/A269M)	---
	Ductile Iron (Spheroidal/Nodular Iron) (ASTM A536)	---
	Copper and Copper Alloy (ASTM B283/B283M)	---
	Copper Alloy (ASTM B584)	---
	Aluminum and Aluminum Alloy (ASTM B209/B209M)	---
	Aluminum-Bronze (ASTM B148)	
	Bronze, Nickel-Aluminum (MIL-B-24480)	
Fasteners	CRES As specified (see 6.2)	---
	---	Nickel-Copper-Aluminum (QQ-N-286) As specified (see 6.2)

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TABLE I. Electro-mechanical valve actuator material requirements - Continued.

Component	Class 1 and 3	Class 2
Mechanical travel stops	CRES	---
	---	Nickel-Copper-Aluminum (QQ-N-286)
Gears, pinions, shafts, and output carrier	CRES	
	Aluminum-Bronze (ASTM B148)	
	Bronze, Nickel-Aluminum (MIL-B-24480)	
	Alloy Steel (ASTM A108)	
Seals and gaskets (for environmental sealing applications)	Buna-N (ASTM D2000 BF, BG, or BK)	
	Ethylene propylene diene terpolymer (EPDM) (ASTM D2000 DA)	
	Fluorocarbon elastomer (FKM) (SAE AMS7276 or SAE AMS7259)	
	Polytetrafluoroethylene (PTFE) (ASTM D4894)	

3.2.1 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Prohibited materials. The valve actuators shall not contain any chemicals categorized as “prohibited” in accordance with T9070-AL-DPC-020/077-2 for NAVSEA programs or NAS411-1 for all other programs.

3.2.3 Electrical. Electrical materials shall be in accordance with MIL-DTL-917.

3.2.4 Lubricants, sealants, and compounds. Lubricants, sealants, and compounds shall be compatible with mating mechanical and electronic equipment.

3.2.5 See-through material. Where used, see-through material (e.g., electronic displays, enclosure applications, light covers, safety guards) shall be transparent (e.g., all visible light is let through). The material shall be waterproof and anti-electrostatic. See-through materials shall not crack, craze, or form any surface irregularities. See-through materials shall be protected from scratches and abrasions. All see-through materials shall be colorless. See-through materials shall be installed with mechanical attachments (e.g., bezels, clips, threads). Substances (e.g., cement, epoxy, adhesive) may be used along with the mechanical attachment.

3.3 Material preparation and conditioning (see 4.5.1).

3.3.1 Forgings. Forging billets shall have been worked down to at least a two to one reduction from a cast ingot. The final forging shall be a totally wrought product, free of any dendritic structure.

3.3.2 Castings. Castings shall be free of adhering sand, hard spots, shrinkage, cracks, or scale. Castings shall be smooth and well cleaned, both internally and externally. All fins and roughness shall be removed. Unacceptable visual surface discontinuities, including tears, inclusions, gas holes, and potentially harmful defects, shall be removed. Unless otherwise specified (see 6.2), castings shall not be repaired, plugged, impregnated, brazed, burned in, or peened.

3.3.3 Surface finishes. All actuator external and internal surface finishes shall be smooth; free of burrs, sharp edges, and irregularities.

3.3.4 Anodizing. Aluminum components exposed to the atmosphere shall be capable of being anodized. Aluminum and aluminum alloys shall have anodic coatings applied and sealed in accordance with MIL-A-8625.

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3.3.5 Priming and painting. Materials that are not corrosion-resistant shall be cleaned and painted in accordance with MIL-DTL-15090, MIL-DTL-24441, or TT-P-1757, or powder coated. Mechanical contact surfaces of supports, items requiring adjustments, internal surfaces, and surfaces prepared for welding shall not be painted unless deemed necessary for operating conditions. Powder coatings shall be in accordance with MIL-PRF-24712.

3.3.6 Heat resistance and encapsulating coatings. If heat resistant coatings are applied, they shall be non-proprietary and permanently bonded. Encapsulating coatings shall not be used on the main enclosure (see 6.3.6) and sub-assemblies.

3.4 Mechanical design (see 4.5.1).

3.4.1 Enclosures. Enclosures shall be in accordance with NEMA type 6P (for submersible enclosures) or type 7 (for explosion-proof enclosures) of ANSI/NEMA 250. Electronic components located within an enclosure shall be in accordance with MIL-DTL-2036. All circuit cards within enclosures shall be spaced so that the heat created by the failure of circuit card components shall not cause failure of other circuit cards in close proximity. External connections shall be accessible to personnel at all times. Any user control located on a controller (see 6.3.3) within an enclosure shall be accessible to personnel at all times, shall be unobstructed, and shall not require enclosure disassembly to gain access. All enclosure assemblies shall be removable, repairable, and replaceable as a complete unit.

3.4.1.1 Main enclosure (required). The main enclosure shall include the following components and sub-assemblies:

a. Output drive sub-assembly (see 6.3.10) connection
b. Motor gear assembly
c. Valve stem primary position measuring device (see 6.3.15)

3.4.1.2 Main enclosure (optional). The following components and sub-assemblies may be included within the main enclosure:

a. Motor (see 6.3.9)
b. Controller
c. Handwheel gear assembly (with integral manual handwheel [see 6.3.5])
d. Torque measuring device

3.4.1.3 Output drive sub-assembly enclosure (required). The output drive sub-assembly enclosure shall be installed in the main enclosure. The output drive sub-assembly enclosure shall include the following components and sub-assemblies:

a. Transmission gears
b. Valve stem connection (see 6.3.14)
c. Output carrier
d. Mechanical valve stem position indicator (see 6.3.8) (if utilized)

3.4.1.4 Output drive sub-assembly enclosure (optional). The following components/sub-assemblies may be included within the output drive sub-assembly enclosure:

a. Handwheel gear assembly (with integral manual handwheel)
b. Mechanical travel stops

3.4.1.5 Controller enclosure. If not installed within the main enclosure, the controller shall have an enclosure. The controller enclosure shall either attach directly to the main enclosure or connect to the main enclosure remotely by external connections.

3.4.1.6 Motor enclosure. If not installed within the main enclosure, the motor shall have an enclosure. The motor enclosure shall attach directly to the main enclosure. The motor enclosure shall be a totally enclosed, non-ventilated type.

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3.4.1.7 Handwheel gear assembly (with integral manual handwheel) enclosure. If not installed within the main enclosure or output drive sub-assembly enclosure, the handwheel gear assembly (with integral manual handwheel) shall have an enclosure. The handwheel gear assembly enclosure shall either attach directly to the main enclosure or attach directly to the output drive sub-assembly enclosure.

3.4.2 Fasteners and mechanical travel stops. Threads for fasteners and mechanical travel stops shall be in accordance with ASME B1.1. Bolts and nuts shall be in accordance with ASME B18.2.1 and ASME B18.2.2. Metric equivalent sizes are acceptable.

3.4.3 Seals and gaskets. Seals and gaskets shall retain their original shape and elasticity. They shall not deform (i.e., take a high compression set resulting in a significant relaxation) over their life expectancy in the application they are being used for in accordance with ASTM D395. The seals and gaskets shall be capable of being properly seated and reseated with ease numerous times, without deformation or material degradation. Seals and gaskets shall be compatible with and prevent the infiltration or leakage of the following:

a. Dust
b. Dirt
c. Oil
d. Steam condensate
e. Naval distillate fuel (F-76, MIL-DTL-16884)
f. Aviation fuel (JP-5, MIL-DTL-5624)
g. Seawater
h. Cleaning solutions
i. Other fluid forms (e.g., condensate, potable water, gray water)
j. Any mixtures of the above

3.4.4 Gearing systems and actuator torque requirements. The actuator shall transmit power from the motor or handwheel to the valve stem through a gearing system. The resultant torque applied to the valve stem connection shall be equal to the required torque value within acceptable tolerances. The required ranges of torque value for valve travel in each direction shall be as specified (see 6.2) and as stated by the valve manufacturer based upon the internal pressure, differential pressure, and frictional loads. Torque values shall be in inch-pound or foot-pound units. An actuator back drive prevention feature shall be incorporated that results in no change of position of the valve from forces applied to the attached valve member (i.e., disc, ball, plug) from the piping system fluid. When a motor sub-assembly (see 6.3.11) is used, there shall be a capability to disconnect the motor sub-assembly's motor output drive from the main enclosure's motor gear assembly. When an actuator motor is installed inside the main enclosure, the motor shall be fully removable for servicing.

3.4.5 Manual actuation. The actuator shall be equipped with a handwheel gear assembly with an integrated manual handwheel to manually actuate the valve. The handwheel shaft shall be of solid construction (e.g., no split shafts) and shall be non-rising. The handwheel shall protrude from the actuator. The handwheel shall be visible and no cover or enclosure shall be used. The handwheel shall be accessible at all times in accordance with MIL-STD-1472. The handwheel assembly shall remain operational during repair and maintenance activities unless the output drive sub-assembly is removed. When specified (see 6.2), requirements for an external mechanical connection (e.g., reach rod) in lieu of a handwheel shall be provided.

3.4.6 Motor position and torque operations. Valve stem travel in both opening and shutting directions shall be stopped when the valve stem reaches the desired setpoint position or when maximum user configured valve stem torque has been achieved. Actuator determination of shut state or open state of a valve shall be presented for approval. The position limit device shall be capable of adjusting the valve stroke such that the valve stem will cycle between any two points within the allowable range of stem travel as determined by the valve design (see 6.2). Mechanical torque devices shall be provided with calibration markings. The travel activated device shall remain in step with the valve position during both electrical and manual actuation of the valve. The accuracy of valve travel for rotational position applications shall be ±½ degree. The accuracy of valve travel for linear position applications shall be ±0.05 inch (1.2 mm). The accuracy of the torque from the output drive sub-assembly shall measure within ±10 percent of whichever is greater of the shut torque or open torque values. Loss of power shall not affect actuator

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handwheel operations from being torque limited.

3.4.6.1 Mechanical travel stops. The actuator shall have user adjustable mechanical stops with locking mechanisms for travel in the fully shut and fully open directions consistent with the stem travel as determined by the valve design (see 6.2).

3.4.7 Local mechanical valve stem position indicator. The actuator shall be provided with a local mechanical position indication device. The mechanical valve stem position indicator shall provide open, shut, and intermediate valve position indications. These valve position indications shall be within ±0.25 inch (6.35 mm) of the actual valve position. The indicator shall be readable during manual actuation.

3.4.8 Weight. The weight of the entire actuator assembly shall not exceed the single person lift requirements of MIL-STD-1472. Alternately, the actuator shall be capable of local disassembly into parts or assemblies that do not individually exceed the single person lift requirements of MIL-STD-1472.

3.4.9 Size. The size envelope for the installation, clearance, and maintenance requirements for the actuator shall be as specified (see 6.2).

3.4.10 Safety. The actuator shall provide protection for personnel against contact with moving parts. The actuator shall provide protection (i.e., insulation) or a local warning (i.e., electronic display or indicator light) for personnel against temperature that will cause harm. The actuator shall have no item protrude or extend (e.g., wire ties) to cause personnel injury.

3.5 Electrical design (see 4.5.1).

3.5.1 Electrical power. The actuator shall be powered by either 115 VAC, 60 hertz, single-phase type 1 power or 440 VAC, 60 hertz, three-phase type 1 power as specified (see 6.2) in accordance with MIL-STD-1399-300-1. All electrical power, regardless of the voltage, shall have its own dedicated cable and connector.

3.5.2 Bonding and grounding. Bonding and grounding shall be in accordance with MIL-DTL-917 and MIL-STD-1310. The actuator shall have an accessible threaded hole exclusively for external bonding and grounding. The actuator shall be provided with details regarding this threaded hole and instructions on how to achieve a proper ground when using it.

3.5.3 Safety. Electrical safety shall be in accordance with MIL-DTL-917 and MIL-STD-1310.

3.5.4 Electrical requirements. Electrical design (e.g., enclosures, components, terminals) shall be in accordance with MIL-DTL-917, MIL-DTL-2036, MIL-DTL-2212, MIL-DTL-32613, and SAE ARP1481. The design or failure of any electrical item shall not prevent manual actuation of the valve.

3.5.4.1 Equipment insulation requirements. Electrical insulation shall comply with the equipment insulation requirements of MIL-DTL-917.

3.5.5 Connection configurations. All internal component wiring connected to component interior surfaces, which cannot be removed and replaced as a complete assembly, shall be terminated, connectorized, and mated to connections on the interior surface of the component. A grouping of internal components that are to be removed and replaced as a complete assembly may be hard-wired together. All internal component wiring that passes through a component surface shall have the opening through which the wiring passes waterproofed. All external component wiring connecting to other component exterior surfaces shall be terminated, connectorized, and mated to connections on the exterior surface of the component. All main enclosure and sub-assembly internal wiring connecting to internal components of the main enclosure and sub-assembly shall be terminated, connectorized, and mated to connections on the exterior surface of the component.

All internal wiring connected to the main enclosure and sub-assembly interior surfaces shall be terminated, connectorized, and mated to connections on the interior surface of the main enclosure or sub-assembly. All external wiring connected to the main enclosure or sub-assembly exterior surfaces shall be terminated, connectorized, and mated to connections on the exterior surface of the main enclosure or sub-assembly. All external wiring shall be terminated by multipin connectors using either contact pins or sockets. Circuitous routing of wire shall be avoided. Connections shall be accessible for users to connect and disconnect with ease.

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3.5.5.1 Network communication. If utilized (see 6.2), the following shall be the only approved methods for network communication:

a. Discrete (hard-wired) contacts

b. Analog 4-20mA current loop

c. Highway addressable remote transducer communication protocol

Upon actuator power-up, the actuator shall automatically connect to the network. If network activity is lost after a network command has been sent to the actuator, the actuator shall continue executing the command. Regardless of the state of network connectivity, all local command, control, and indications shall function.

3.5.6 Connectors. Connectors shall be in accordance with MIL-C-24368, MIL-DTL-917, MIL-DTL-26482, MIL-STD-1399-502, or SAE AS50151, as appropriate. No tools shall be required for connecting or removing connectors.

3.5.6.1 Main enclosure. The main enclosure shall have the following provisions for connections:

a. Valve stem primary position measuring device connection

b. Torque measuring device connection

c. Actuator internal temperature sensor connection

d. Motor thermal protection monitoring connection (if the motor is housed in the main enclosure)

e. User controls for valve operations and network connectivity (if utilized)

f. Electronic display user controls for programming (if utilized)

g. Network communication connection (if utilized)

h. External programming device connection (if utilized)

i. External diagnostic device connection (if utilized)

j. Hard-wired remote indicator light device connection (if utilized)

k. Hard-wired remote manual control and indicator light device connection (if utilized)

l. Motor power connection (if the motor is housed in the main enclosure)

m. Controller power connection (if the controller is housed in the main enclosure)

3.5.6.2 Controller. The controller shall have the following provisions for connections:

a. Controller power connection

b. Motor power connection

c. Valve stem primary position measuring device connection

d. Actuator internal temperature sensor connection

e. Motor thermal protection monitoring connection

f. User controls for valve operations and network connectivity (if utilized)

g. Torque measuring device connection

h. Network communication connection (if utilized)

i. External programming device connection (if utilized)

j. External diagnostic device connection (if utilized)

k. Electronic display user controls for programming (if utilized)

l. Hard-wired remote indicator light device connection (if utilized)

m. Hard-wired remote manual control and indicator light device connection (if utilized)

3.5.7 Controller. The controller shall be powered by 115 VAC, 60 hertz, single-phase type 1 power or 440 VAC, 60 hertz, three-phase type 1 power as specified (see 6.2). The controller shall provide power to the following:

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a. Motor

b. Indicator lights

c. Valve stem primary position measuring device

d. Torque measuring device

e. Electronic display (if utilized)

f. Hard-wired remote indicator light device (if utilized)

g. Hard-wired remote manual control and indicator light device (if utilized)

3.5.7.1 Controller function. The controller shall have inputs and outputs that provide monitoring, status, and control of the actuator.

3.5.7.1.1 Inputs.

a. Valve position. The controller shall use information provided by the valve stem primary position measuring device.

b. Torque value. The controller shall use torque value information provided by the torque measuring device or internal controller torque software logic.

c. Actuator internal temperatures. The controller shall use temperature information provided by the actuator internal temperature sensors.

d. Motor winding temperature. The controller shall use the thermal protection monitoring of the motor to obtain motor winding temperature rise values.

3.5.7.1.2 Outputs. The controller shall enable the following motor functions:

a. Phase power on for clockwise rotation (e.g., valve shut)

b. Phase power off for clockwise rotation (e.g., valve shut)

c. Phase power on for counterclockwise rotation (e.g., valve open)

d. Phase power off for counterclockwise rotation (e.g., valve open)

e. Speed control (see 3.6.2.3)

3.5.7.2 Cooldown. If utilized (see 6.2), the actuator controller may have a cooldown capability. The time required for cooldown shall be determined by the manufacturer of the actuator with the constraint that the maximum allowable cooldown time shall not exceed one duty cycle. The cooldown capability shall be activated when any one of the following conditions occur:

a. Upon completion of two complete open-shut cycles or two complete shut-open cycles (see 6.3.4)

b. When operating actuator exceeds allowable temperature requirements

c. Upon exposure to temperatures that will damage the actuator, or its components as determined by any internal or external temperature sensor(s)

All existing valve commands shall be completed before cooldown is activated. The cooldown capability shall restart if needed during an active cooldown event. During cooldown, motor activation shall be prohibited but all other functions shall remain operational. There shall be a capability to override the cooldown feature to resume operation. The time remaining in cooldown to resume operation shall be calculated. If the actuator is furnished with an electronic display, then the actuator shall display the remaining cooldown time on that display when in that mode. If the actuator is not furnished with an electronic display, the existing actuator indicator light(s) shall be used to serve as a local indicator for when the actuator is in a cooldown period.

3.5.7.3 Actuator torque and position control. The controller shall be configured to operate for position and torque seating applications as determined by the valve seating method selection. The controller shall prohibit the actuator from exceeding the shut torque value and open torque value of the user configurable operational settings. The controller shall prohibit the actuator from exceeding the shut limit and open limit of the user configurable operational settings. The controller shall use the shut limit setpoint to command the motor to stop when the valve seating method is set to shut limit. The controller shall use the shut torque value to command the motor to stop

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when the valve seating method is set to shut torque. The controller shall use the open limit setpoint to command the motor to stop when the valve seating method is set to shut limit or shut torque. The controller shall use the shut limit and open limit to relate the travel scale to 0 and 100 percent, respectively. The controller shall use the travel scale to calculate the valve position in whole percent increments.

3.5.7.4 Fault condition. The actuator controller shall process each of the following fault conditions:

a. Valve position information not present: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

b. Torque information not present: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

c. Internal temperature threshold value exceeded: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

d. Motor winding temperature rise threshold value exceeded: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

e. Valve position information does not match commanded direction: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

f. Valve position information does not change after a predetermined time: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

g. Valve commanded position not reached after a predetermined time: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

h. In 440 VAC, 60 hertz three-phase applications, if a phase is not present: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

i. In 440 VAC, 60 hertz three-phase applications, if phases are reversed: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

j. Power out of range: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

k. Controller fails to operate properly: the controller shall remove power from the motor and all required indications and messaging shall be enabled.

No fault condition shall cause damage to the actuator. The actuator shall be restored to normal operations using local user controls. Actuators with built-in electronics shall retain the last five fault condition occurrences. These last five fault conditions shall be accessible locally through an electronic display or remotely via a machinery control system network connection or an external diagnostic device. In the event of an actuator power loss, the last five fault conditions shall be retained and accessible upon actuator restoration.

3.5.8 Motor. The motor shall be powered by 115 VAC, 60 hertz, single-phase type 1 power or 440 VAC, 60 hertz, three-phase type 1 power (see 6.2). The motor shall meet service A requirements in accordance with MIL-DTL-17060 for integral horsepower and service A requirements in accordance with MIL-M-17059 for fractional horsepower. The motor shall be capable of rotational change within 1 second. Motor winding temperature rise shall not exceed the thermal rating of the motor insulation system (see 4.5.12). No batteries or backup circuitry shall be incorporated into the motor.

3.5.8.1 Duty cycle. Duty of motors shall be continuous, intermittent, varying, or short-time, as specified (see 6.2). Motors of other than continuous duty shall have a defined duty cycle in accordance with MIL-DTL-17060 or MIL-M-17059, as applicable.

3.5.8.2 Loss of phase power. Loss of phase power monitoring shall be integrated with the controller.

3.5.8.3 Thermal protection monitoring. Thermal protection monitoring of the motor winding temperature rise shall be integrated with the controller.

3.5.8.4 Sealed insulation system. Motors shall be provided with sealed insulation systems in accordance with MIL-DTL-17060 or MIL-M-17059.

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3.5.9 Indicator lights. Indicator lights shall be used to provide actuator status to the user. When three-phase power is used, the indicator lights shall not use the same phase power as the controller. The controller shall enable or disable the indicator lights. The indicator lights shall always be viewable to personnel, shall be unobstructed, and shall not be enclosed unless see-through materials are used. The indicator lights shall be in accordance with MIL-DTL-3661 and MIL-STD-1472. The indicator lights shall be removable and replaceable.

3.5.10 Electronic display. When the actuator is furnished with a user display screen, the display (e.g., Liquid Crystal Display [LCD], Light Emitting Diode [LED]) shall provide user configurable operational setting capabilities, actuator status, and fault conditions to the user. Information entered, or pre-determined options selected on the display, shall be accomplished by user controls. The display shall rotate in 90-degree increments to accommodate viewing of the screen in various positions. The display shall always be illuminated. The display shall not be susceptible to screen burn-in. The display shall not use a screen saver. The display shall be in accordance with MIL-STD-1472. The display shall be removable and replaceable.

3.5.10.1 Electronic display control. If utilized (see 6.2), the electronic display shall have a setup screen and an operational screen. A means shall be provided to toggle between the setup screen and the operational screen, and vice versa. The setup screen shall provide the user with the capability to enter user configurable operational settings. When setup is complete, a confirmation message shall be displayed and the setup screen shall change to the operational screen. If setup is not complete, the operational screen shall not be accessible. If the setup screen is accessed after setup is complete, the operational screen shall display automatically after a period of 5 minutes of inactivity. The operational screen shall display messages for the operational status and fault conditions of the actuator. The fault condition messages shall scroll across the bottom of the screen.

3.5.10.2 Electronic display messaging. Messages shall be in the form of alphanumeric text, symbols, icons, or characters. All messages shall be displayed on the electronic display, an external programming device, or an external diagnostic device. Where network communications are utilized, all messages shall be made available to network communications. All message formats shall be in accordance with MIL-STD-1472.

3.5.10.2.1 Electronic display setup messaging. Upon completion of entering the user configurable operational settings, the following message shall be displayed: “SETUP COMPLETE”.

3.5.10.2.2 Electronic display operational messaging. For valve operations, messaging shall be in accordance with the following requirements.

3.5.10.2.2.1 Actuator position messaging. When the valve is equal to or greater than the shut limit and less than the shut setpoint, a message shall indicate “SHUT”. When the valve is equal to or greater than the shut setpoint and less than or equal to the open setpoint, a message shall indicate “XX % OPEN” in 1-percent increments, where XX represents a whole numerical value from 1 to 99. When the valve is greater than the open setpoint and less than or equal to the open limit, a message shall indicate “OPEN”.

3.5.10.2.2.2 Actuator position over-ride messaging. If the valve is stopped locally and the valve is equal to or greater than the shut limit and less than the shut setpoint, a message shall indicate “STOPPED LOCALLY SHUT”. If the valve is stopped remotely and the valve is equal to or greater than the shut limit and less than the shut setpoint, a message shall indicate “STOPPED REMOTELY SHUT”. If the valve is stopped locally and the valve is equal to or greater than the shut setpoint and less than or equal to the open setpoint, a message shall indicate “STOPPED LOCALLY XX % OPEN”. If the valve is stopped remotely and the valve is equal to or greater than the shut setpoint and less than or equal to the open setpoint, a message shall indicate “STOPPED REMOTELY XX % OPEN”. If the valve is stopped locally and the valve is greater than the open setpoint and less than or equal to the open limit, a message shall indicate “STOPPED LOCALLY OPEN”. If the valve is stopped remotely and the valve is greater than the open setpoint and less than or equal to the open limit, a message shall indicate “STOPPED REMOTELY OPEN”.

3.5.10.2.2.3 Actuator cooldown cycle messaging. If utilized (see 6.2), during actuator cooldown, a message shall indicate “COOLDOWN #### SEC REMAINING” in 1-second increments, where #### represents whole seconds remaining until the cooldown period has ended. Each of the above messages (see 3.5.10.2.2.1 and 3.5.10.2.2.2) shall be displayed accordingly at all times during the cooldown cycle.

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3.5.10.2.2.4 Network status messaging. When network connectivity is utilized, a message shall always be displayed indicating the state of the connection. When remotely connected, a message shall indicate “REMOTE CONNECTION”. When not remotely connected, a message shall indicate “LOCAL ONLY”. When the actuator connects to more than one network channel, a message shall display the status of each network channel.

3.5.10.2.2.5 Fault condition messaging. The following messages shall be displayed for each fault condition:

a. “VALVE POSITION INFORMATION NOT PRESENT”
b. “TORQUE INFORMATION NOT PRESENT”
c. “INTERNAL TEMPERATURE EXCEEDED”
d. “MOTOR TEMPERATURE EXCEEDED”
e. “VALVE POSITION DOES NOT MATCH COMMANDED DIRECTION”
f. “VALVE POSITION DID NOT CHANGE”
g. “VALVE POSITION NOT REACHED”
h. “PHASE NOT PRESENT” (if required)
i. “PHASES REVERSED” (if required)
j. “POWER OUT OF RANGE”
k. “CONTROLLER FAILURE”

The message for each fault condition shall repeat, with a 1-second delay between iterations, until the fault condition no longer exists. While a fault condition message is active, if additional fault conditions occur, each additional fault condition shall have its message added to the current fault condition or conditions message sequence. There shall be a 1-second delay between each fault condition message.

3.5.11 User controls. The main enclosure controller or the hard-wired remote manual control and indicator light device shall each have user controls. These user controls shall perform the following actions:

a. Valve seating. Method to seat valve (shut limit or shut torque)
b. Shut limit. The setpoint that defines when the valve is at its maximum shut travel
c. Shut torque. The maximum torque value allowed inside the shut zone
d. Shut setpoint. The location where the valve is deemed shut (e.g., hard seated)
e. Open limit. The setpoint that defines when the valve is at its maximum open travel
f. Open torque. The maximum torque value allowed inside the open zone
g. Open setpoint. The location where the valve is deemed open (e.g., full travel)
h. Valve travel speed. The unconstrained valve travel speed (see 3.6.2.3c and 6.3.13)
i. Network communication. To select network communication on or off (if utilized)
j. Network address. To enter a network address (if utilized)

The user configurable operational settings shall be entered in the sequence as listed. Upon completion of the user configurable operational settings, the controller shall enable all required indications and messaging. Each user configurable operational setting shall prohibit an incorrect data entry, invalid setting, or make another user configurable operational setting invalid. Valid user configurable operational settings shall never result in prohibiting actuator operations. Each user configurable operational setting shall be retained upon loss of power. Upon replacement of any component or device, the user configurable operational settings shall be confirmed or modified starting in the sequence as listed.

3.5.12 User commands. User commands to the actuator shall be provided from at least one of the following locations:

a. User controls located on the main enclosure
b. User controls located on the controller

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c. Network communication

d. Hard-wired remote manual control and indicator light device

e. External programming device

f. External diagnostic device

Access to the actuator from the above for setup, programming, and diagnostics shall be password protected. The user interface shall have the ability to be programmed for a period of inactivity to lock the screen and require the user to re-enter the password.

3.5.13 Electromagnetic environment. Unless otherwise specified (see 6.2), the actuator shall be in accordance with DOD-STD-1399-070-1 and MIL-STD-461 using the limits for metallic ships below deck.

3.6 Operation. The actuator shall operate, without repair or adjustment, under the following conditions (see 4.5.13):

a. 20,000 cycles using motor operation

b. 1,000 transfers between manual to motor operation

c. 1,000 transfers between motor to manual operation

d. In any installed position or orientation

3.6.1 Actuator manual operation. The valve shall close by clockwise rotation of the handwheel when facing the handwheel. The direction of movement for closing and opening the valve shall be indicated by arrows on the face of the handwheel rim. The maximum tangential force applied to operate the handwheel shall be 40 pounds regardless of the diameter of the handwheel. The handwheel assembly shall meet the following requirements (see 4.5.13 and 4.5.18):

a. The handwheel assembly shall be operational whether the actuator is energized or not.

b. User activation of the handwheel shall override motor valve stem operations.

c. The handwheel assembly shall have complete control of valve stem operations within 1 second upon activation of the handwheel by a user.

d. The handwheel shall not spin while the motor is operating.

e. The handwheel assembly shall not add resistance to the operation of the motor.

f. The failure of electrical components shall not prevent manual actuation of the valve via the handwheel assembly.

g. The motor shall not engage or add resistance to the operation of the handwheel gear assembly.

h. Upon de-activation of the handwheel gear assembly, the motor shall fully engage and take control of actuator operations.

3.6.2 Actuator electrical operation. The controller shall enable the following motor functions (see 4.5.13 and 4.5.18):

a. Power for valve shut rotation in one direction

b. Power for valve open rotation in the opposite direction

c. Speed control

3.6.2.1 Actuator operation – shut-to-open-shut movement. Upon receipt of a local or remote command to open the valve, the controller shall provide the necessary power signal for the actuator motor to move the valve stem in the open direction. Upon receipt of a local or remote command to shut the valve, the controller shall provide the necessary power signal for the actuator motor to move the valve stem in the shut direction. Upon receipt of a local or remote command to move a valve, the controller shall provide the necessary power signal to start moving the valve stem within 1 second. Upon receipt of a local or remote command to stop a valve, the controller shall stop providing a power signal to move the valve stem within 1 second. When the commanded position is achieved, the associated controller power signal shall cease.

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3.6.2.2 Actuator operation – time to process commands. When the actuator has completed the valve commanded position and receives a command to move the valve, the motor shall move the valve within 1 second. While the actuator is moving and receives a new command to move the valve, the motor shall move the valve according to the new command within 1 second. All initial motor movements shall gradually increase in speed.

3.6.2.3 Actuator operation – speed control. The actuator controller shall adjust the motor speed when starting and when stopping or when approaching the shut or open setpoints. The speed of the motor shall be managed upon receipt of a command as follows:

a. When the valve position originates at the shut limit or between the shut limit and the shut setpoint, the motor shall have its speed increased gradually until the valve position is greater than the shut setpoint. When the valve is commanded to terminate between the open setpoint and the open limit or at the open limit, the motor shall have its speed decreased gradually until the valve is at the commanded position.

b. When the valve position originates at the open limit or between the open limit and the open setpoint, the motor shall have its speed increased gradually until the valve position is greater than the open setpoint. When the valve is commanded to terminate between the shut setpoint and the shut limit or at the shut limit, the motor shall have its speed decreased gradually until the valve is at the commanded position.

c. When the valve position is greater than the shut setpoint and less than the open setpoint, the motor shall operate at the valve travel speed. When the valve position originates at a location greater than the shut setpoint and less than the open setpoint, the motor shall have its speed increased gradually until the valve travel speed is achieved. When the valve position terminates at a location greater than the shut setpoint and less than the open setpoint, the motor shall have its speed decreased gradually until the valve is at the commanded position.

3.6.3 Actuator operating time. For manual or motor operation, the actuator shall be capable of an actuator cycle (see 6.3.2) within the operating times as specified for manual operation and for motor operation. For manual operating times not specified (see 6.2), for fully opening or fully shutting a valve, the operating time shall be within five times the motor operating time. For motor operating times not specified (see 6.2), the actuator shall operate with 10 to 12 inches of linear stroke per minute or 5 seconds per inch of valve bore (rotational travel) (see 4.5.13 and 4.5.18).

3.6.4 Over-travel. If mechanical stops are installed, the mechanical stops shall prevent over-travel of the actuator without damage to any components (see 4.5.13 and 4.5.18).

3.6.5 Indicator light operation. The controller shall enable indicator lights to illuminate solid. The controller shall enable indicator lights to flash in accordance with the flash coding requirements of MIL-STD-1472. The duration of the flash shall be in accordance with MIL-STD-1472 (see 4.5.13 and 4.5.18). There shall be two flash coding capabilities:

a. Standard flash coding: an ongoing flash at the same continuous rate

b. Unique flash coding: an ongoing flash with a specific rate assigned to a fault condition

3.6.5.1 Shut indicator light operation. The shut indicator light shall illuminate red. Upon completion of the user configurable operational settings, the shut indicator light shall illuminate three times using standard flash coding. These three illuminations shall be synchronized with the open indicator light.

a. During valve opening (e.g., full travel), the shut indicator light shall illuminate as follows:

(1) Illuminate solid upon receipt of the command to open the valve

(2) Illuminate solid when the valve is in motion and on the shut limit or in motion between the shut limit and the open setpoint

(3) Illuminate solid when the valve is stopped on the shut limit, stopped between the shut limit and the open setpoint, or stopped on the open setpoint

(4) Not illuminate when the valve is in motion and has passed the open setpoint, stopped on the open limit, or stopped between the open setpoint and the open limit

b. During valve shutting (e.g., hard seated), the shut indicator light shall illuminate as follows:

(1) Illuminate using standard flash coding upon receipt of the command to shut the valve

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(2) Illuminate using standard flash coding when the valve is in motion and on the open limit, or in motion between the open limit and the shut limit

(3) Not illuminate when the valve is stopped between the open limit and the open setpoint

(4) Illuminate solid when the valve is stopped at the open setpoint, stopped between the open setpoint and the shut limit, or stopped on the shut limit

c. Upon failure of the controller, the shut indicator light shall illuminate solid until normal operations resume.

3.6.5.2 Open indicator light operation. The open indicator light shall illuminate green. Upon completion of the user configurable operational settings, the open indicator light shall illuminate three times using standard flash coding. These three illuminations shall be synchronized with the shut indicator light.

a. During valve opening (e.g., full travel), the open indicator light shall illuminate as follows:

(1) Illuminate using standard flash coding upon receipt of the command to open the valve

(2) Illuminate using standard flash coding when the valve is in motion and on the shut limit, or in motion between the shut limit and the open limit

(3) Not illuminate when the valve is stopped between the shut limit and the shut setpoint

(4) Illuminate solid when the valve is stopped on the shut setpoint, stopped between the shut setpoint and the open limit, or stopped on the open limit

b. During valve shutting (e.g., hard seated), the open indicator light shall illuminate as follows:

(1) Illuminate solid upon receipt of the command to shut the valve

(2) Illuminate solid when the valve is in motion and on the open limit, or in motion between the open limit and the shut setpoint

(3) Illuminate solid when the valve is stopped on the open limit, stopped between the open limit and the shut setpoint, or stopped on the shut setpoint

(4) Not illuminate when the valve is in motion and has passed the shut setpoint, stopped on the shut limit, or stopped between the shut setpoint and the shut limit

c. Upon failure of the controller, the open indicator light shall illuminate solid until normal operations resume.

3.6.5.3 Cooldown indication. During actuator cooldown, both the open and shut indicator lights shall illuminate using standard flash coding until the cooldown period has ended. Both lights shall be synchronized.

3.6.5.4 Network connection indicator light operation. If the actuator is configured for network connection, a network indicator light shall be required. The network indicator light shall illuminate white. Upon receipt of a local actuator controller command by a user to enable network connectivity (i.e., pushbutton, switch, electronic display) and after the connection of the controller to the network, the network indicator light shall illuminate solid. Upon receipt of a local actuator controller command by a user to disable network connectivity (i.e., pushbutton, switch, electronic display) and after the disconnection of the controller from the network, the network indicator light shall extinguish. Upon loss of an established controller network connection, the network indicator light shall flash. After 30 consecutive seconds of controller network connection loss, the network indicator light shall extinguish. While establishing a controller network connection, the network indicator light shall flash. The network controller shall establish network connection within 30 consecutive seconds. Once a controller has established network connection, the network indicator light shall illuminate solid.

3.6.5.5 Fault indicator light operation. The fault indicator light shall illuminate amber. When a fault condition exists, and an electronic display is not used, the fault indicator light shall illuminate using unique flash coding for each fault condition. The unique flash coding for each fault condition shall repeat, with a 1-second delay between iterations, until the fault condition no longer exists. While a fault condition unique flash coding is active, if additional fault conditions occur, each additional fault condition shall have its unique flash coding added to the current fault condition or conditions flash coding sequence. There shall be a 1-second delay between each fault condition unique flash coding. A unique flash coding shall be created for each of the following fault conditions:

a. Valve position information not present

b. Torque information not present

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c. Internal temperature threshold value exceeded

d. Motor winding temperature rise threshold value exceeded

e. Valve position information does not match commanded direction

f. Valve position information does not change after a predetermined time

g. Valve commanded position not reached after a predetermined time

h. Phase not present

i. Phases reversed

j. Power out of range

k. Controller fails to operate properly

When a fault condition exists, and an electronic display is used, the fault indicator light shall illuminate using standard flash coding until the fault condition no longer exists. Upon failure of the controller, the fault indicator light shall illuminate solid until normal operations resume.

3.7 Environmental conditions.

3.7.1 Temperature. The actuator shall operate under a temperature range of -30 to 65 °C (-20 to 150 °F) (see 4.5.9 and 4.5.10). Heat generated inside the actuator shall not damage or cause failure of the actuator or its components. The use of mechanical or fluid cooling is prohibited. The actuator shall be designed with a surface area to dissipate heat via ambient air, heat sinks, or similar method.

3.7.2 Shock. The actuator shall meet grade A shock requirements of MIL-DTL-901 for class 1, type A, deck mounted equipment with unrestricted mounting orientation (see 4.5.14).

3.7.3 Vibration. The actuator shall comply with the vibration requirements in accordance with MIL-STD-167-1 without damage, deformation, cracking, or loosening of fasteners (see 4.5.15).

3.7.4 Airborne noise. Unless otherwise specified (see 6.2), the actuator shall be in accordance with airborne sound acceptance criteria of the shipboard equipment noise requirements of MIL-STD-1474 (see 4.5.11).

3.7.5 Humidity. Unless otherwise specified (see 6.2), the actuator shall operate at relative humidity up to 100 percent, including conditions where condensation takes place in and on the actuator (see 4.5.5).

3.7.6 Submersion. Service 1 and 3 actuators shall meet submersible enclosure requirements in accordance with MIL-DTL-917 and MIL-DTL-2036 (see 4.5.6).

3.7.7 Explosion proof. Service 2 and 3 actuators shall meet explosion proof requirements (i.e., design, construction, and testing) in accordance with MIL-DTL-2036 (see 4.5.7).

3.7.8 Fire resistance. When subjected to the fire resistance test of 4.5.8, the actuator shall be capable of manual operation with the full shut and full open torque values falling within the acceptable tolerances of the required torque value of the actuator.

3.8 Cybersecurity. Actuators and their controllers shall meet the cybersecurity requirements of MIL-DTL-32613.

3.9 Installation. The actuator shall be capable of being installed in any position or orientation. Installation shall be in accordance with MIL-STD-1472. Any change in actuator orientation with respect to the valve shall be accomplished with a minimum of disassembly or field modification.

3.10 Maintenance, repair, and replacement. The actuator shall be designed for ease of maintenance, repair, and replacement actions in accordance with MIL-STD-1472. If applicable, the use of lifting lugs, alignment guides, access panels, and any other features to facilitate the ease of maintenance, repair, and replacement shall be incorporated. The actuator shall be designed to the maximum extent possible to facilitate maintenance, repair, and replacement actions while the actuator is mounted to a valve. The actuator shall be designed so that maintenance, repair, and replacement actions are performed at the lowest maintenance level (see 6.3.7).

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3.11 Interchangeability. All actuator sub-assemblies, parts, and components performing the same function shall be interchangeable without modification or rework. All actuator sub-assemblies, parts, and components shall be designed to prevent incorrect installation. When interchanged, all actuator sub-assemblies, parts, and components shall operate as designed in their original intended role without modification or rework. All actuator sub-assemblies, parts, and components that are replacements shall have the same form, fit, and function as the original sub-assemblies, parts, and components. All actuator sub-assemblies, parts, and components that are replacements shall be of the same accreditation, in accordance with this specification, as the original sub-assemblies, parts, and components that are being superseded or replaced.

3.12 Special tools. Special tools shall be defined as tools not listed in the GSA Global Supply Catalog. When specified (see 6.2), the following special tools shall be provided:

a. External programming device
b. External diagnostic device
c. Printed circuit board diagnostic and repair device
d. Portable toolbox to house special and common tools

3.12.1 External programming device. The external programming device shall authenticate both the device and the user before the actuator accepts any commands from external devices. The external device shall be password protected. After a period of 5 minutes of inactivity, the user shall have to re-enter the password. If the device is de-energized, the user shall have to re-enter the password when the device is re-energized.

3.12.2 External diagnostic device. The external diagnostic device shall authenticate both the device and the user before the actuator accepts any commands from external devices. The external device shall be password protected. After a period of 5 minutes of inactivity, the user shall have to re-enter the password. If the device is de-energized, the user shall have to re-enter the password when the device is re-energized.

3.13 Identification plate. Unless otherwise specified (see 6.2), actuators shall be provided with an identification plate that shall contain the following information:

a. Manufacturer’s name and internet address
b. Manufacturer’s drawing number
c. Model number
d. Serial number
e. Firmware version number (if applicable)
f. Control software version number (if applicable)
g. Torque rating
h. Actuator voltage, amperage, frequency, and number of phases
i. Weight

The identification plate shall be affixed to the actuator when delivered.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2)
b. Conformance inspection (see 4.3)

4.2 First article inspection. First article inspection shall be conducted on the lowest quantity of actuators possible when a first article sample is required (see 3.1). All examinations and tests specified in 4.5 shall be conducted without repair or adjustment during the test. All examinations and tests shall be conducted concurrently if practicable. Noncompliance with any requirement or the presence of one or more defects shall constitute cause for rejection. Any redesign or modification of the actuator to comply with requirements, or any redesign or modification following failure to meet requirements, shall be re-examined and re-tested.

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4.3 Conformance inspection. Conformance inspection shall be conducted on actuators as manufactured under the contract. Each actuator as manufactured shall be examined and tested as specified in 4.5.1 and 4.5.18. All examinations and tests shall be conducted without repair or adjustment during the test. All examination and tests shall be conducted concurrently if practicable. Noncompliance with any requirement or the presence of one or more defects shall constitute cause for rejection. Any redesign or modification of the actuator to comply with requirements, or any redesign or modification following failure to meet requirements, shall be re-examined and re-tested.

TABLE II. First article and conformance inspections.

Characteristic	First article	Conformance	Requirement	Verification
Material	X	X	3.2 through 3.3.6	4.5.1
Mechanical design	X	X	3.4 through 3.4.10	4.5.1
Electrical design	X	X	3.5 through 3.5.13	4.5.1
Prohibited materials	X	--	3.2.2	4.4
Dielectric withstanding voltage	X	--	3.5.4.1	4.5.2
Insulation resistance	X	--	3.5.4.1	4.5.3
Electromagnetic interference	X	--	3.5.13	4.5.4
Stalled motor heat rise	X	--	3.5.8	4.5.12
High environmental temperature operation	X	--	3.7.1	4.5.9
Low environmental temperature operation	X	--	3.7.1	4.5.10
Airborne noise	X	--	3.7.4	4.5.11
Humidity	X	--	3.7.5	4.5.5
Enclosure (submersion)	X	--	3.7.6	4.5.6
Enclosure (explosion proof)	X	--	3.7.7	4.5.7
Fire	X	--	3.7.8	4.5.8
Endurance and operation	X	--	3.4.4, 3.6, 3.6.1, 3.6.2, 3.6.3, 3.6.4, and 3.6.5 through 3.6.5.4	4.5.13
Shock	X	--	3.7.2	4.5.14
Vibration	X	--	3.7.3	4.5.15
Overload	X	--	3.6.1	4.5.16
Fault conditions	X	--	3.5.10.2.2.5 and 3.6.5.5	4.5.17
Operational compliance and adjustment	--	X	3.6.1, 3.6.2, 3.6.3, 3.6.4 and 3.6.5 through 3.6.5.4	4.5.18

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4.4 Prohibited materials. Prohibited materials shall be verified as specified (see 3.2.2 and 6.2).

4.5 Test, analysis, demonstration, and examination.

4.5.1 Visual examination. The actuator shall be visually examined to determine conformance with the material, dimensional, functional, and design requirements of 3.2 through 3.5.13. This examination shall include a review of actuator design specifications to ensure functional capabilities comply with the requirements of 3.2 through 3.5.13. Actuator covers and access panels shall be removed or opened as necessary to conduct the examination. Permanently sealed components shall be visually examined prior to assembly.

4.5.2 Dielectric withstanding voltage test. The actuator shall be subjected to the dielectric withstanding voltage test in accordance with MIL-DTL-917.

4.5.3 Insulation resistance test. The actuator shall be subjected to the insulation resistance test in accordance with MIL-DTL-917. The test criteria shall be in accordance with the equipment insulation requirements of MIL-DTL-917.

4.5.4 Electromagnetic environment. The actuator shall be subjected to the electromagnetic interference test in accordance with DOD-STD-1399-070-1 and MIL-STD-461.

4.5.5 Humidity test. The actuator shall be subjected to the humidity test in accordance with method 507 of MIL-STD-810.

4.5.6 Enclosure test (submersion). Service 1 and 3 actuators shall be subjected to tests for protection against ingress of water (temporary submersion) in accordance with ANSI/NEMA 250.

4.5.7 Enclosure test (explosion proof). Service 2 and 3 actuators shall be subjected to explosion-proof testing conducted in accordance with the explosion-proof verification requirements of MIL-DTL-2036.

4.5.8 Fire test. The actuator shall be subjected to a fire test in accordance with the following requirements. The actuator shall not be protected with insulation material of any form during testing, except where such protection is part of the design of the actuator sub-assemblies, parts, and components. Any test equipment used shall not affect the test results. The actuator shall be tested so that the valve stem connection is at the lowest point. Prior to subjecting the actuator to fire, three complete manual actuator cycles shall be conducted. The actuator shall be completely engulfed in fire at a minimum temperature of 1,800 °F (983 °C) for a continuous period of 30 minutes. After the continuous period of 30 minutes, the fire shall be extinguished. Three complete manual actuator cycles shall be conducted 30 minutes after the fire is extinguished. The full shut and full open torque values shall be recorded for each of the three manual actuator cycles. Test acceptance criteria shall be that all recorded torque values are within acceptable tolerances of the required torque value of the actuator. This test does not apply to a remote mounted controller.

4.5.8.1 Fire testing provisions. All fire tests specified in this document shall be conducted by an independent testing laboratory that is accredited to ISO/IEC 17025 and is approved by the Technical Authority (see 6.3.12). Accreditation shall be obtained from a recognized accreditation body such as American Association for Laboratory Accreditation (A2LA) (www.A2LA.org) or International Code Council's International Accreditation Services (IAS) (www.IASonline.org). The scope of accreditation shall include specific flammability and fire tests. All other fire test provisions shall be as specified (see 6.2 and 6.4).

4.5.9 High temperature test. All temperature values used in this test shall be stabilized at the required temperature (successive temperatures not varying more than ±1 °C [1.8 °F] over a period of 30 minutes). The actuator shall be exposed to the high non-operating design temperature of 65 °C (150 °F) for a period of 4 hours. The high non-operating temperature shall then be lowered to the high operating design temperature and maintained. While maintaining the high operating design temperature, the actuator shall be functionally tested using three motor operation actuator cycles. The temperature shall then be lowered to the ambient temperature and the actuator shall be functionally tested using three motor operation actuator cycles. Failure of the actuator to operate as defined at the high operating design temperature or the ambient temperature shall constitute cause for rejection.

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4.5.10 Low temperature test. All temperature values used in this test shall be stabilized at the required temperature (successive temperatures not varying more than ±1 °C [1.8 °F] over a period of 30 minutes). The actuator shall be exposed to the low non-operating design temperature of -20 °C (20 °F) for a period of 1 hour. The low non-operating temperature shall then be raised to the low operating design temperature and maintained. While maintaining the low operating design temperature, the actuator shall be functionally tested using three motor operation actuator cycles. The temperature shall then be raised to the ambient temperature and the actuator shall be functionally tested using three motor operation actuator cycles. Failure of the actuator to operate as defined at the low operating design temperature or the ambient temperature shall constitute cause for rejection.

4.5.11 Airborne noise test. This test shall be performed in accordance with shipboard equipment noise requirements of MIL-STD-1474.

4.5.12 Stalled motor heat rise test. The stalled motor heat rise test shall be conducted as follows:

a. All associated faults shall be deactivated

b. De-energize the actuator and secure the motor shaft to prevent rotation in any direction

c. Energize the actuator and command the actuator to move

d. The actuator shall operate for the equivalent run time of one-half duty cycle

e. De-energize the actuator and measure the motor winding temperature in accordance with the temperature measurements requirements of MIL-DTL-917

f. The motor winding temperature shall not exceed the rating of the motor

4.5.13 Endurance and operation test. The actuator shall be subjected to an endurance test and an operational test to determine conformance with the requirements specified in 3.4.4 and 3.6 through 3.6.5.4. These tests may occur separately or simultaneously as warranted.

4.5.13.1 Initial conditions.

a. Power – Minimum steady user voltage and maximum steady state frequency in accordance with DOD-STD-1399-300-1.

b. Loading – Actuator installed on the largest size valve (gate or globe [type 1] or quarter-turn [type 2]) for which the actuator is designed, with valve pressurized at system normal operational pressure (see 6.2) or 125 pounds per square inch gauge (psig) if no pressure is specified, or with a simulated load that equals the pressurized valve conditions.

c. Communications – If utilized, a simulated network connection to the actuator shall be provided only for the operation test portion.

4.5.13.2 Test requirements. The actuator shall be electrically cycled (shut-open-shut) 10,000 times with the stem in the vertical position, 5000 times with the stem in the horizontal position, and 5000 times with the stem in the inverted vertical position. After every 200 cycles, the actuator valve shall be manually cycled (shut-open-shut) to determine compliance with 3.6.1. The electric actuating time for an opening or shutting cycle shall be recorded at the completion of every 200th cycle and shall be as specified in 3.6.2. The actuator electrical operation maximum torque and running torque shall be measured and recorded at every 200th cycle to verify compliance in accordance with the torque range per 3.4.4. Electrical operation shall be as specified in 3.6.2. Over-travel prevention (via mechanical stop performance) shall be as specified by 3.6.4. The test fixture shall not provide any rotational restraint to the actuator during the test in order to ensure that the requirements of the anti-rotation device are satisfied. Indicator light performance, including network connection, if utilized, shall be as specified by 3.6.5 through 3.6.5.4.

4.5.14 Shock test. The actuator shall be subjected to shock tests in accordance with MIL-DTL-901.

4.5.15 Vibration test. The actuator shall be subjected to vibration tests in accordance with MIL-STD-167-1.

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4.5.16 Overload test. An overload test shall be conducted with the actuator installed on the largest size of valve (gate or globe [type 1] or quarter-turn [type 2]) for which the actuator is designed, with the valve pressurized at system normal operational pressure (see 6.2) or 125 psig if no pressure is specified. The valve stem and the actuator output drive bushings shall be lubricated, and the valve packing gland shall be unloaded. If mechanical stops are applicable, they shall be properly set up and utilized during this test.

a. With the torque limit device and the position limit device bypassed, the valve shall be electrically cycled shut until the motor stalls, at which time the valve shall be manually opened using the handwheel assembly. The torque required to open the valve shall not exceed that allowed per 3.6.1.

b. With the torque limit device and the position limit device bypassed, the valve shall be electrically cycled open and back seated until the motor stalls, at which time the valve shall be manually opened using the handwheel assembly. The torque required to open the valve shall not exceed that allowed per 3.6.1.

4.5.17 Fault conditions test. The actuator shall be subjected to the fault conditions tests of 4.5.17.1 through 4.5.17.11 to determine conformance with the fault light indication requirements specified by 3.6.5.5 and, if utilized, the messaging requirements specified by 3.5.10.2.2.5.

4.5.17.1 Valve position information not present fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The valve position information connection to the controller shall be halted while the actuator is operating. The actuator motor shall stop operations once it has detected the loss of valve position information. The required indications and messages shall also be activated.

4.5.17.2 Torque information not present fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The torque information connection to the controller shall be halted while the actuator is operating. The actuator motor shall stop operations once it has detected the loss of torque information. The required indications and messages shall also be activated.

4.5.17.3 Internal temperature threshold value exceeded fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The internal temperature of the actuator shall be increased beyond the allowable threshold while the actuator is operating. The actuator motor shall stop operations once it has detected the temperature is beyond the allowable threshold (i.e., via internal or externals sensor) except if the actuator utilizes a cooldown period. The required indications and messages shall also be activated. The internal temperature of the actuator shall be decreased beyond the allowable threshold while the actuator is operating. The actuator motor shall stop operations once it has detected the temperature is beyond the allowable threshold. The required indications and messages shall also be activated.

4.5.17.4 Motor winding temperature rise threshold value exceeded fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The motor winding temperature shall be increased beyond the allowable threshold while the actuator is operating. The actuator motor shall stop operations once it has detected the motor winding temperature is beyond the allowable threshold. The required indications and messages shall also be activated.

4.5.17.5 Valve position information does not match commanded direction fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The valve position information shall operate opposite of the actuator commanded position while the actuator is operating. The actuator motor shall stop operations once it has detected the valve position information does not match the commanded direction. The required indications and messages shall also be activated.

4.5.17.6 Valve position information does not change after a predetermined time fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The valve position information shall be stopped from changing while the actuator is operating. The actuator motor shall stop operations once it has detected the valve position information has not changed after a predetermined time. The required indications and messages shall also be activated.

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4.5.17.7 Valve commanded position not reached after a predetermined time fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The valve position information shall be slowed down while the actuator is operating. The actuator controller shall stop motor operations once valve position information has been determined to not be operating quick enough in accordance with the actuator speed setting. The required indications and messages shall also be activated.

4.5.17.8 Phase not present fault condition test. This test (applies to only 440 VAC, 60 hertz three-phase applications) shall verify that the actuator controller de-energizes (when applicable) and the indicator lights are all illuminated (when applicable). This test shall also verify that power to the motor and all of the required indications and messages are enabled accordingly. Each phase shall be individually removed from the actuator while it is operating. The required indications and messages shall also be activated.

4.5.17.9 Phases reversed fault condition test. This test (applies to only 440 VAC, 60 hertz three-phase applications) shall verify that the actuator controller operates accordingly, power is removed from the motor, and all of the required indications and messages are enabled. All possible phase combinations shall be reversed with the actuator de-energized. The actuator shall then be energized with a reversed phase combination. The actuator controller shall detect the phase reversal. The actuator motor shall not be allowed to operate when there is a phase reversal detected by the controller. The required indications and messages shall also be activated.

4.5.17.10 Power out of range fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The power to the actuator shall be decreased to just below the lowest allowable power threshold while the actuator is operating. The actuator motor shall stop operations once it has detected this decrease of power. The required indications and messages shall also be activated. The power to the actuator shall be increased to just above the highest allowable power threshold while the actuator is operating. The actuator motor shall stop operations once it has detected this increase of power. The required indications and messages shall also be activated.

4.5.17.11 Controller fails to operate properly fault condition test. This test shall verify that the actuator controller removes power from the motor and all of the required indications and messages are enabled. The controller processing shall be interrupted while the actuator is operating by simulating a controller fault. The actuator controller shall stop motor operations once it has detected a controller fault. The required indications and messages shall also be activated.

4.5.18 Operational compliance and adjustment. This test shall be performed on each actuator to verify proper functioning of each actuator and to support adjustments (see 6.2). With initial conditions of 4.5.13.1 established (if actuator is installed on a sample valve, the valve stem shall be in the vertical position), the actuator shall be cycled shut-open-shut alternately five times electrically and five times manually. During this test, adjustments shall be made to position indicators, torque and travel activated devices, and mechanical stops. The test fixture shall not provide any rotational restraint to the actuator during the test in order to ensure that the requirements of the anti-rotation device are satisfied. The following shall be verified:

a. Manual operation is as specified in 3.6.1.
b. Electrical operation is as specified in 3.6.2.
c. Operation times are as specified in 3.6.3.
d. Mechanical stops are adjusted to prevent over-travel as specified in 3.6.4.
e. Indicator lights operate as specified in 3.6.5 through 3.6.5.4.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The valve actuator described by this specification is intended for use in naval shipboard propulsion plant and auxiliary system service. This specification is intended to be used in conjunction with a valve specification or drawing for an identified system application. Actuators manufactured to DOD-V-24657(SH) dated 22 July 1985 are allowed when used to replace DOD-V-24657(SH) actuators on legacy ships and when specified by contract.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Actuator type, class, service, and travel (see 1.2).

c. When first article inspection is required (see 3.1).

d. Fastener materials for external bolting of actuator to valve (see 3.2).

e. Requirement for prohibited materials verification (see 3.2.2 and 4.4).

f. When a casting is to be repaired, plugged, impregnated, brazed, burned in, or peened (see 3.3.2).

g. Ranges of torque value for valve travel in each direction (including breakaway torques for OPEN/SHUT as well as running torque) (see 3.4.4).

h. When an external mechanical connection (e.g., a reach rod) in lieu of a handwheel is required (see 3.4.5).

i. Allowable range of valve stem travel (see 3.4.6 and 3.4.6.1).

j. Installation, clearance, and maintenance requirements of the size envelope for the actuator (see 3.4.9).

k. Actuator, controller, and motor power requirements (see 3.5.1, 3.5.7, and 3.5.8).

l. Network communications requirement (see 3.5.5.1).

m. Cooldown requirements and messaging (see 3.5.7.2 and 3.5.10.2.2.3).

n. Duty cycle of motor (see 3.5.8.1 and 6.3.4).

o. Limits and criteria for EMI in accordance with MIL-STD-461 if other than the limits for metallic ships below deck (see 3.5.13).

p. Actuator manual operating time requirement (see 3.6.3).

q. Actuator motor operating time requirement (see 3.6.3).

r. Maximum allowable airborne noise level (see 3.7.4).

s. Operational relative humidity percentage (see 3.7.5).

t. Special tools required (see 3.12).

u. Identification plate information (see 3.13).

v. Additional fire testing provisions (see 4.5.8.1 and 6.4).

w. Valve size, type, body material, seating method, stem size, stem travel, stem material, stem thread requirements, and normal operating pressure (see 4.5.13.1, 4.5.16, and 4.5.18).

x. Packaging requirements (see 5.1).

6.3 Definitions.

6.3.1 Actuator. A system that is comprised of a main enclosure, sub-assemblies, and components (internal and external) that operate a valve to a desired position.

6.3.2 Actuator cycle. The complete movement of the valve from fully shut to fully open back to fully shut or vice versa.

6.3.3 Controller. Comprised of components that provide actuator control (e.g., local and remote) and indications.

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6.3.4 Duty cycle. Unless otherwise specified (see 6.2), the required actuator non-stop operational run time for completion of two open-shut cycles or two shut-open cycles.

6.3.5 Integral manual handwheel. Provides manual movement to operate the actuator and operates independent of the motor.

6.3.6 Main enclosure. Houses internal components and provides mounting options for external sub-assemblies and components that are required for operation of the actuator.

6.3.7 Maintenance levels. OPNAVINST 4700.7 defines three levels of maintenance echelons as organizational-level (O-level), intermediate-level (I-level), and depot-level (D-level) in which O-level is considered the lowest maintenance level.

6.3.8 Mechanical valve stem position indicator. Provides local actuator status of attached valve's linear or rotational position.

6.3.9 Motor. Provides torque to operate the actuator.

6.3.10 Output drive sub-assembly. Enables actuator motor driven or manual operation of the valve stem. Also, the location where additional torque output gear ratios may be housed. This sub-assembly may also be used for actuator to valve mounting or for attaching applicable external sub-assemblies and components.

6.3.11 Sub-assembly. An attachment that houses components that is externally attached or remotely connected.

6.3.12 Technical authority. The technical authority has the authority, responsibility, and accountability to establish, monitor, and approve technical standards, tools, and processes in conformance to higher authority policy, requirements, architectures, and standards. NAVSEA 05S may be contacted to determine the appropriate technical authority for any given requirement related to equipment and systems of U.S. Navy ships and submarines.

6.3.13 Unconstrained valve travel speed. Unmodified electric actuator motor output speed exerted onto the attached valve.

6.3.14 Valve stem connection. The interface between the actuator's output drive sub-assembly and the valve stem.

6.3.15 Valve stem primary position measuring device. Provides linear or rotational valve position information to the actuator controller.

6.4 Additional fire testing provisions. The Technical Authority reserves the right to witness the tests and perform any of the tests set forth herein where such testing is deemed necessary to assure compliance to prescribed requirements. A2LA (www.A2LA.org) or International Code Council's IAS (www.IASonline.org) maintain a full list of the approved laboratories that are accredited to ISO/IEC 17025 and provide details on how to obtain and maintain accreditation (see 6.2).

6.5 Classification cross-referencing.

	DOD-V-24657	MIL-DTL-24657A
Type code	I	1
	II	2
Class code	I	1
	II	2
	III	3
Service code	I	1
	II	2
	III	3
Travel code	I	1
	II	2

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6.6 Subject term (key word) listing.

Linear

Mechanical indication

Multi-turn

Network interface

Operator

Quarter-turn

Remote control

Valve operation

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Preparing activity:
Navy – SH
(Project 4810-2019-001)

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